

Cyrus Guided Notes: Stanford CS229 Machine Learning

Stanford course pack

2026-05-15

Course source

Official source: <https://cs229.stanford.edu/>

Why this course is in the system. Empirical risk, gradient descent, generative models, SVMs, neural networks, and generalization.

Learning output. Next batch target for a machine-learning formula and problem route.

Prerequisite checkpoint

- Calculus
- Linear algebra
- Probability

Formula checkpoint

Do not memorize these first. Read each formula as a sentence: what changes, what is observed, what is optimized, and what output it produces.

$$\begin{aligned}\min_{\theta} \frac{1}{N} \sum_{i=1}^N \ell(f_{\theta}(\mathbf{x}_i), y_i) + \lambda \Omega(\theta) \\ \theta \leftarrow \theta - \alpha \nabla_{\theta} J(\theta) \\ p(y = 1 | \mathbf{x}) = \sigma(\theta^{\top} \mathbf{x}) \\ p(\mathbf{x} | y) = \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_y, \Sigma_y)\end{aligned}$$

Visual page

Draw data, model, loss surface, gradient step, and validation error.

GoodNotes pages

- Page ST-ML001: empirical risk

- Page ST-ML002: gradient descent
- Page ST-ML003: probabilistic model

Web interaction

A loss-landscape slider that changes learning rate and overfitting.

Obsidian and Notion

Layer	Output
Obsidian	Stanford ML -> CS229
Notion	Problem-set status and formula mastery.

First study move

Open the official source, copy the prerequisite checkpoint into GoodNotes, then write one formula in your own words before watching or reading more.